connect 12V AC secondary of transformer X1 to the PCB. Connect 230V AC, switch S2, primary of X1, CT1 and the appliance externally using suitable wires.

Once the circuit assembly is done, write/burn applianceguard.hex file to the MCU using a suitable programmer. Then, mount the MCU on the PCB. It is recommended to use a 28-pin IC base for the MCU. Keep VR1 and VR2 to the minimum and VR3 to the maximum positions.

Now, pass the phase/line wire of the appliance through CT1 and measure the current using a clamp meter, as explained in Calibration section below. Using S2, switch on the circuit and the appliance. Relay RL1 will get energised with a small delay and AC appliance will switch on.

Calibration

Make sure that one end of CT1 as well as VR1 is grounded. Only one line of AC power, phase or live (L) should pass through the current coil.

AC current value is displayed on the first row of LCD1. Initially, value displayed on LCD1 may not be correct. Adjust VR1 slowly till LCD1 shows the value read on the clamp meter.

Adjust VR2 till voltage at pin 24 of IC1 is low (0V) and adjust VR3 till voltage at pin 25 of IC1 is high (5V). Remove the clamp meter after adjustment and calibration.

To adjust the high or maximum limit to trip the relay, adjust VR3 to the required maximum current limit (say, 1.5 times the working current value displayed in the first row). Similarly, to adjust the low or minimum limit to trip the relay, adjust VR2 to the required minimum current limit (say, 0.5 times the working current). Now, the circuit is ready to use.

Once the value crosses the high or low limit, the buzzer will start beeping and the corresponding LED (LED1 or LED2) will start blinking. If blinking continues for more than 10 seconds, the corresponding LED will blink quickly and the relay will switch off, thus protecting the appliance or equipment. Press reset switch S1 to restart the program.

Note. Current transformers are available with various capacities like 10A, 30A, etc, which indicates the maximum current these can measure correctly. Select one with more than the required capacity—preferably double (or more) of the working current range. (Current program can read maximum 50A.)

Caution. Care should be taken while handling 230V AC power supply. EFY

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